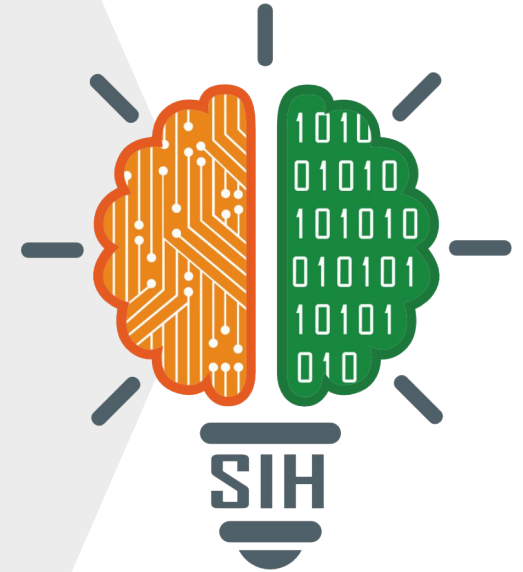


SMART INDIA HACKATHON 2025



TITLE PAGE

- **Problem Statement ID –** SIH25013
- **Problem Statement Title -** Real Time Public Transport Tracking for Small Cities
- **Theme-** Transportation & Logistics
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal) –** Code Crew



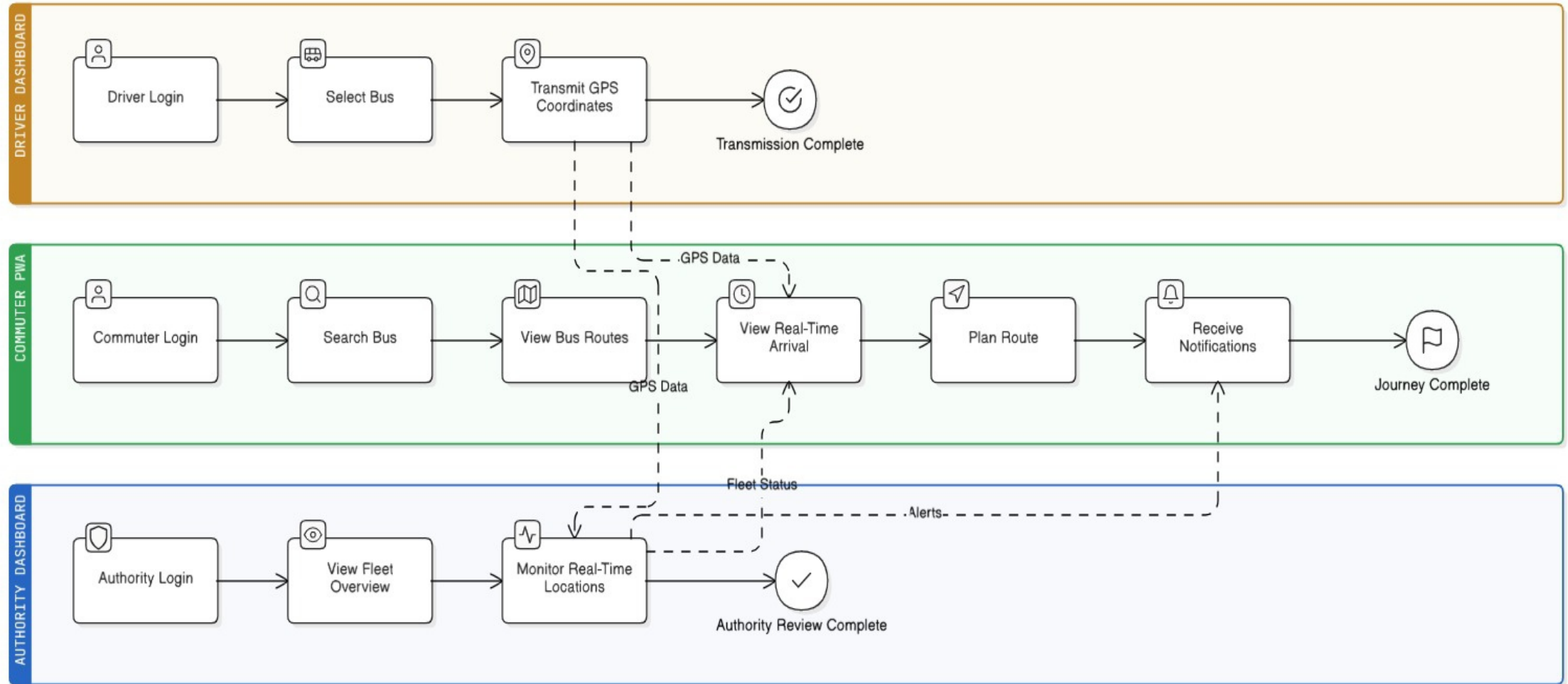
IDEA / SOLUTION :

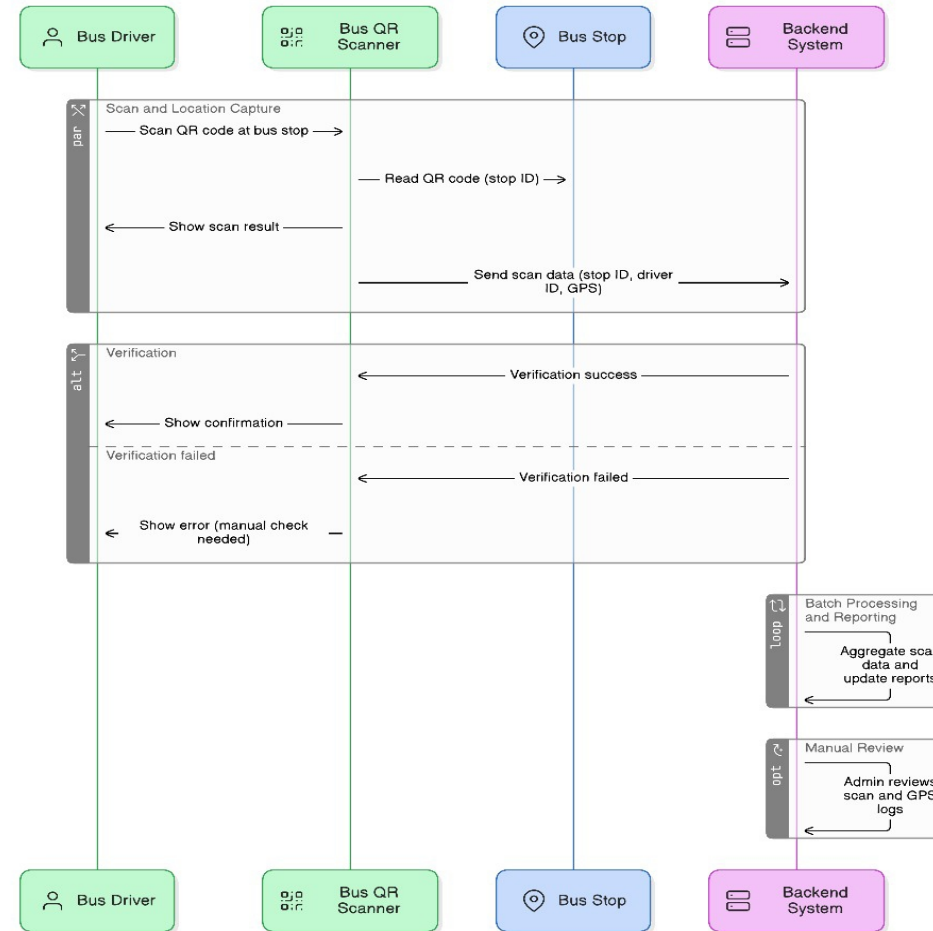
Implementation of a web platform integrating **GPS-based** real time tracking of buses

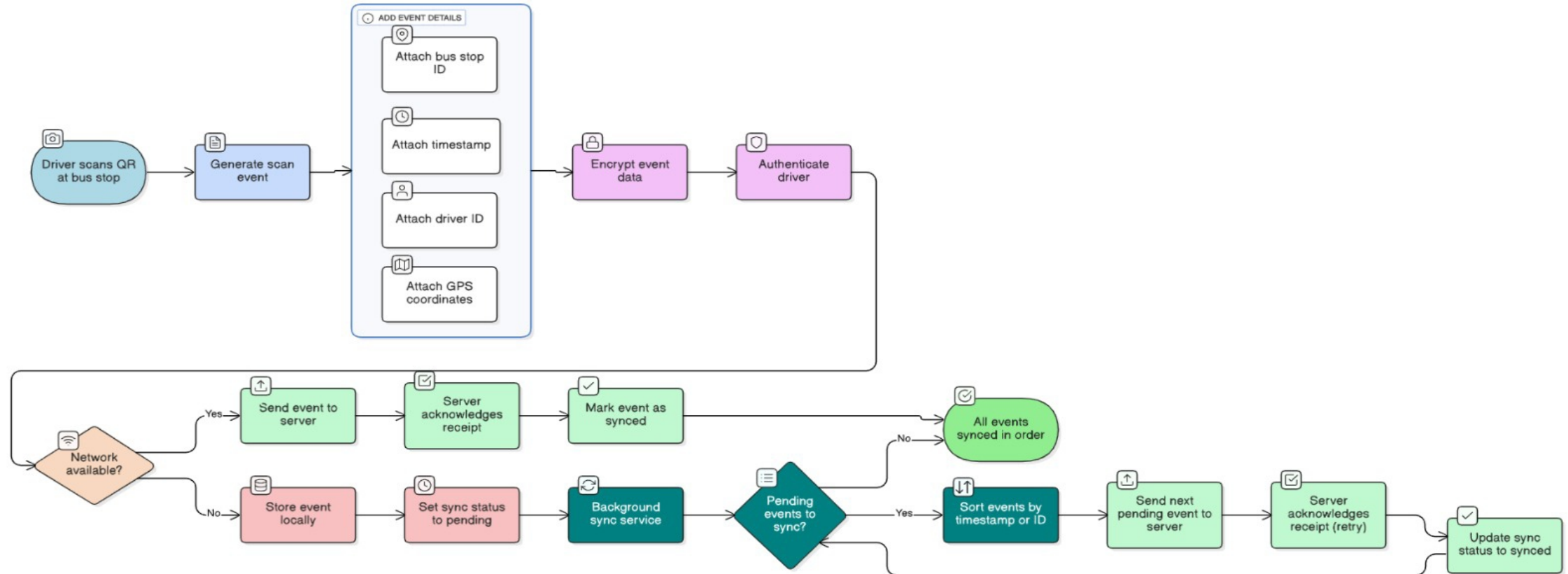
- **Live Tracking** : See buses in real time on the map
- **Accurate ETAs** : Get arrival times for any stop
- **Stop Verification** : QR based verification
- **GPS Cross-Check** : Confirms location accuracy
- **Route Info** – View routes, stops and schedules
- **Search & Filter** – Quickly find the right bus

PROBLEM RESOLUTION :

- ❖ **GPS based tracking** provides real-time bus locations, accurate ETAs, and automatic route verification for passengers and operators.
- ❖ **QR scanning** at each stop verifies bus arrivals, cross-checks GPS data, and updates the server in real time.
- ❖ Store Scan feature with **IndexedDB** saves QR scans offline in the browser and syncs them later, ensuring **data reliability**.







Frontend:

- **HTML, CSS, JavaScript, Tailwind** – Core technologies used for developing the UI of the website.
- **Firebase Authentication** – Secure driver login
- **Leaflet.js** – interactive maps (buses, stops, routes)

Backend + Database :

- **Firebase Firestore** – real-time bus locations, routes, and QR scan data
- **Firebase Functions** – verify QR scans against GPS, calculate ETAs, send updates
- **Firebase Offline Mode** – acts as **Store Scan**
- **IndexedDB** – offline store scans

(Self Hosted Database is recommended)



Technical Feasibility

- Uses proven technologies – Leaflet, Firebase/Node.js + MongoDB
- QR codes are easy to generate and scan
- GPS provides accurate location data for real-time tracking
- Offline storage via IndexedDB ensures no data loss

Economic Feasibility

- Low-cost implementation
- Minimal hardware requirement (driver smartphone + QR stickers)
- No need for expensive map APIs using Leaflet

“The solution is technically simple, low-cost and operationally practical making it feasible and scalable for real-world use.”

Challenges

- **Driver Compliance** – QR scan depends on drivers scanning at every stop
- **Offline Data Management** – Handling scans when network is poor

Passengers

Real-time bus locations reduce waiting time and uncertainty
Accurate ETAs and Easy route information helps plan trip efficiently

Bus Operators / Authorities

Better monitoring of bus locations and adherence to schedule
Reduced chances of missed stops or route deviations
Improved decision-making through real-time and historical data

City / Transit System

Enhanced reliability and trust in public transport
Data-driven insights for future route planning and optimization

“The system improves passenger experience, supports drivers, and enables operators to monitor, optimize, and enhance overall public transport reliability.”

CODE
CREW



THANK YOU